



IMDB Movie Recommendation System Using Storylines

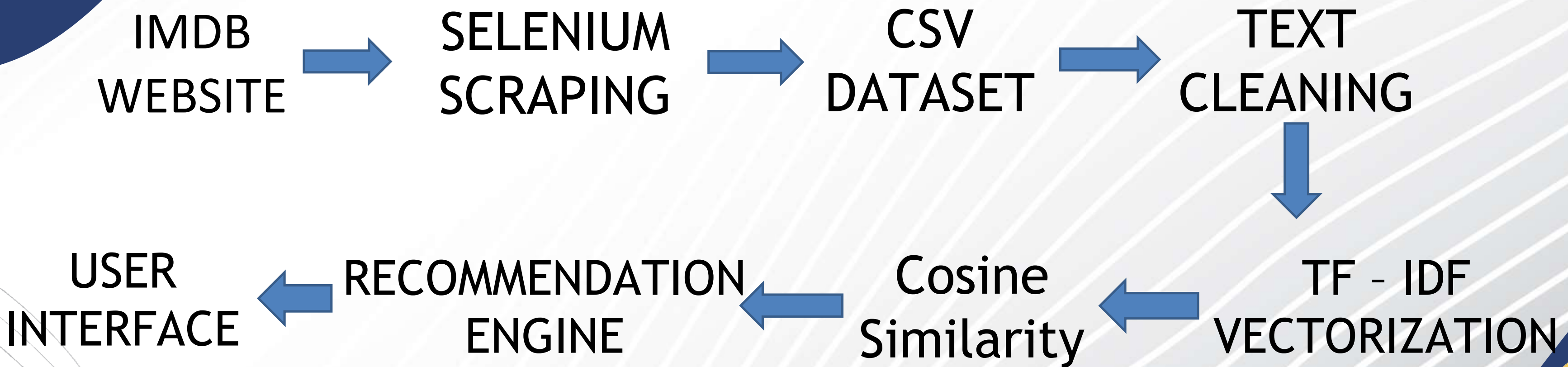
INTRODUCTION

1. Developed a content-based Movie Recommendation System using storyline similarity instead of user ratings.
2. Collected movie data (titles & storylines) from IMDb using Selenium and stored it in CSV format.
3. Applied NLP techniques, TF-IDF vectorization, and Cosine Similarity to analyze and compare movie storylines.
4. Built an interactive web application using Stream lit to recommend the top 5 similar movies instantly.

Tools and Technology Used

- **Python:** Core programming language used to build and integrate the entire recommendation system.
- **Selenium:** Automates browser actions to scrape movie data from IMDb.
- **Pandas:** Handles data storage, cleaning, and manipulation using structured Data Frames.
- **NLTK:** Performs text preprocessing like stop word removal and text cleaning for NLP tasks.
- **Scikit-learn:** Provides machine learning tools for TF-IDF vectorization and similarity computation.
- **TF-IDF:** Converts textual storylines into numerical vectors by measuring word importance.
- **Cosine Similarity:** Calculates similarity between movie storylines to generate recommendations.
- **Streamlit:** Builds an interactive web interface to display real-time movie recommendations.

SYSTEM ARCHITECTURE



OBJECTIVE

1. **Scrape IMDb movie data using Selenium:** Automatically collects movie names and storylines from IMDb using browser automation.
2. **Perform text preprocessing using NLP:** Cleans and prepares storyline text by removing unwanted words, symbols, and formatting.
3. **Convert text into numerical vectors using TF-IDF:** Transforms textual data into numerical form so machines can understand and process it.
4. **Calculate similarity using Cosine Similarity:** Measures how similar different movie storylines are to generate recommendations.
5. **Develop an interactive web interface using Streamlit:** Creates a user-friendly web app using Streamlit for real-time movie recommendations.

FUTURE SCOPE

1. Enhance recommendation accuracy using advanced NLP models like Word2Vec or BERT.
2. Implement hybrid recommendation system by combining content-based and collaborative filtering.
3. Integrate real-time APIs (like IMDb or TMDB) for updated movie data.
4. Improve user interface by adding movie posters, ratings, and personalized user profiles.

The background features a light gray field with a series of thin, white, wavy lines that create a sense of motion. In the corners, there are decorative elements: the top-left and bottom-right corners have bold, overlapping wavy bands in dark blue, gold, and black; the top-right and bottom-left corners contain faint, thin-lined grid patterns.

THANK YOU